

Engine Only Operates in Idle Mode

Symptoms

- The engine will **only** operate in idle mode.

How To Use This Tree

This symptom tree can be used to troubleshoot engine idle symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

None.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.)	
	STEP 1A. Check the idle/rated switch SIGNAL wire for short circuit.	Greater than 100k ohms?
STEP 2.	Check the the engine harness to the C.I.B. cable.	
	STEP 2A. Check the idle/rated switch SIGNAL wire.	Less than 10 ohms?

STEP 1. Check the C.I.B. wiring.

STEP 1A. Check the idle/rated switch SIGNAL wire for short circuit.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step
Check idle/rated switch SIGNAL wire. • Disconnect the idle/rated switch SIGNAL wire from the X1 connector. • Place one test lead on the idle/rated switch SIGNAL pin in connector C1. • Place the other test lead on the battery 1 voltage return terminal of the cascade module.	Greater than 100k ohms? YES Repair : Replace the wire(s). Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Greater than 100k ohms? NO	2A

STEP 2. Check the engine harness to the C.I.B. cable.

STEP 2A. Check the idle/rated switch SIGNAL wire.

Conditions :

- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C4 from the engine harness.

Action	Specification/Repair	Next Step
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Check the idle/rated switch SIGNAL wire. • Place one test lead in the idle/rated switch SIGNAL pin of the C1 connector. • Place the other test lead on another pin in the C1 connector. • Repeat for all other pins in the C1 connector.	Less than 10 ohms? YES Repair : Troubleshoot the appropriate fault code. Reference the Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF; or the ISM and QSM 11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF; or the X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346, Section TF; or the equipment manufacturer service information.	Repair complete
	Less than 10 ohms? NO Repair : Replace the cable.	Repair complete